

SRUJANA G

AI Engineer | GenAI | LLM | RAG | AI Agents | Python

+91 9141363498 | srujana.ganesh21@gmail.com | LinkedIn | GitHub

Professional Summary

AI Engineer with production experience building and deploying GenAI applications, LLM-powered services, RAG pipelines, and agentic AI workflows at Zynthora.ai. Hands-on experience developing LLM-integrated microservices, hybrid retrieval systems, vector and knowledge-graph architectures, and production inference pipelines using Python, FastAPI, vLLM, Redis, Docker, and GPU infrastructure. Built multi-agent RAG systems using ReAct, HyDE, and Corrective RAG, and worked with vector databases including Qdrant and pgvector. Experienced in taking AI systems from experimentation through deployment, optimization, validation, and production readiness.

Professional Experience

Associate AI Engineer - Zynthora.ai

Sept 2025 – Present

- Develop and productionize AI/GenAI solutions across Healthcare, Telecom, BFSI, and Defence use cases, contributing across model development, AI service integration, deployment, optimization, and production readiness.

OneVoice - Real-Time Meeting Translation Platform (ASR, MT, TTS, Meeting Minutes, RAG Side Panel)

- Developed production AI services integrating ASR, machine translation, TTS, meeting summarization, and RAG-based contextual assistance into a real-time meeting platform.
- Built a streaming chunking service that combines fragmented ASR output using word/sentence boundaries and a rolling Redis context buffer before passing coherent text to downstream translation services.
- Integrated M2M-100 with NVIDIA Triton Inference Server over gRPC, implementing KV-cache decoding and dynamic batching for efficient GPU inference.

ReSolveRCM - AI-Driven Healthcare Claims Denial Management

- Built and deployed an isolated LLM-powered AI microservice using vLLM and Mistral-7B-Instruct, exposing a structured JSON API for automated denial root-cause analysis across 10 categories and 8 recommended actions.
- Developed a hybrid RAG pipeline combining retrieval and cross-encoder reranking, achieving **(85%+ Recall@5, 88%+ accuracy)**.
- Designed the RAG retrieval layer using vector search and cross-encoder reranking to retrieve relevant denial knowledge before LLM-based root-cause classification and recommended-action generation.

Unified Speech-to-Speech (S2S)

- Implemented the speech encoding pipeline using mHuBERT, extracting 768-dimensional contextual speech representations from input audio for downstream discrete speech-unit modeling.
- Developed the continuous-to-discrete speech representation stage using MiniBatchKMeans, training a 1,000-cluster codebook to quantize mHuBERT embeddings into discrete speech units.
- Validated the encoder and quantization pipeline on the English-Hindi S2S system, ensuring consistent discrete-unit representations for downstream speech translation components.

PRISM - Social Sentiment Intelligence Platform

- Designed a multi-stage NLP pipeline combining preprocessing, heuristic filtering, LLM-based sentiment classification, crisis-severity detection, and impact scoring for high-volume social content.
- Implemented credibility, bot, and promotional-content scoring to reduce noisy or low-value social signals before downstream sentiment and crisis analysis.
- Built LLM-driven classification workflows for sentiment and crisis severity, converting unstructured social-media content into structured intelligence for downstream analytics.

AI Engineer Intern - AIRAT Systems, NIT Karnataka

Jan 2025 – May 2025

- Designed to support surgical training by offering performance scores, suggestions, and visual feedback to help surgeons refine their skills.
- The system processes laparoscopic surgery videos to detect and track instrument movements using computer vision. A reinforcement learning model evaluates tool paths and provides real-time feedback for improving precision and reducing errors.

Personal Projects

CogniGraph - Agentic Hybrid GraphRAG Platform

- Built an agentic RAG system combining a **Neo4j** knowledge graph, vector index, and a multi-agent **ReAct** loop (Planner, Graph, Vector, Critic, Synthesis) with **HyDE** retrieval and **Corrective RAG (CRAG)** quality gating, backed by semantic chunking and LLM-driven entity extraction across 7 Dockerized services.

Education

Sahyadri College of Engineering and Management

2021 – 2025

B.E., Artificial Intelligence and Machine Learning | CGPA: **8.83** / 10

Technical Skills

Languages, Frameworks & Speech AI: Python, FastAPI, Pydantic, PyTorch, LangChain, Whisper ASR, M2M-100, Chatterbox TTS, VAD
LLMs & Generative AI: Gemini, Claude, Groq/Llama-3, vLLM, Mistral-7B, LoRA, RAG, HyDE, CRAG, ReAct, Reinforcement Learning, OpenCV
Data, Vector/Graph DBs & DevOps: Qdrant, pgvector, Neo4j, Redis, MongoDB, PostgreSQL, Docker, CUDA/GPU, Celery, Git